



Outbreak Investigation and Surveillance



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Lecture Objective

- ▶ At the end of this unit students must be able to :-
 - ▶ Discuss reasons to investigate an outbreak
 - ▶ Differentiate the different patterns of outbreaks
 - ▶ Describe key steps of outbreak investigations

- ▶ Define public health surveillance
- ▶ Differentiate types of surveillance
- ▶ Comprehend features of good surveillance
- ▶ Evaluate a surveillance system

Brainstorming

Define these words

- ▶ Epidemic/ Outbreak
- ▶ Endemic
- ▶ Hyper endemic
- ▶ Pandemic
- ▶ Sporadic



Why to Investigate an Outbreak

1. To control and prevent spread of disease
2. Opportunity for research
3. Training opportunity
4. Opportunity for program evaluation
5. Prevent future outbreaks
6. Improve surveillance and outbreak detection



Patterns of Epidemics

There are three principal types Epidemics:-

1. Common Source Epidemics
2. Propagated Epidemics
3. Mixed Epidemics



1. Common Source Epidemics

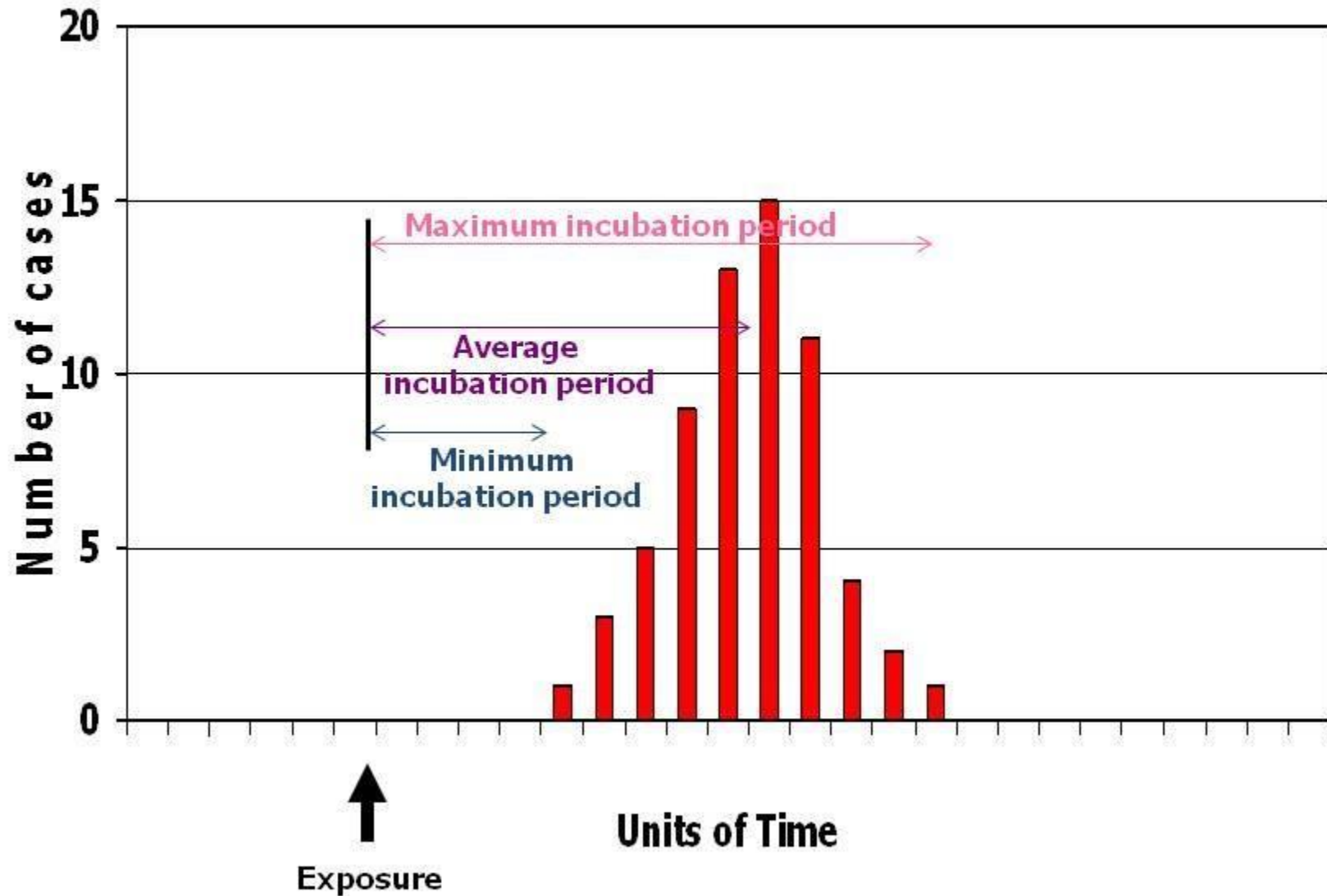
- The disease arise from a single, shared or 'common' source, such as a batch of bad food, industrial pollution, contaminated water.
- Similar incubation period
- A rapid rise and fall of an epidemic curve
- Controlling the source stops the outbreak

a) Point Source Epidemics

- Within one incubation period, suggesting that cases did not arise from person-to-person spread
- The outbreak was of short duration
- Exposure that did not persist over time



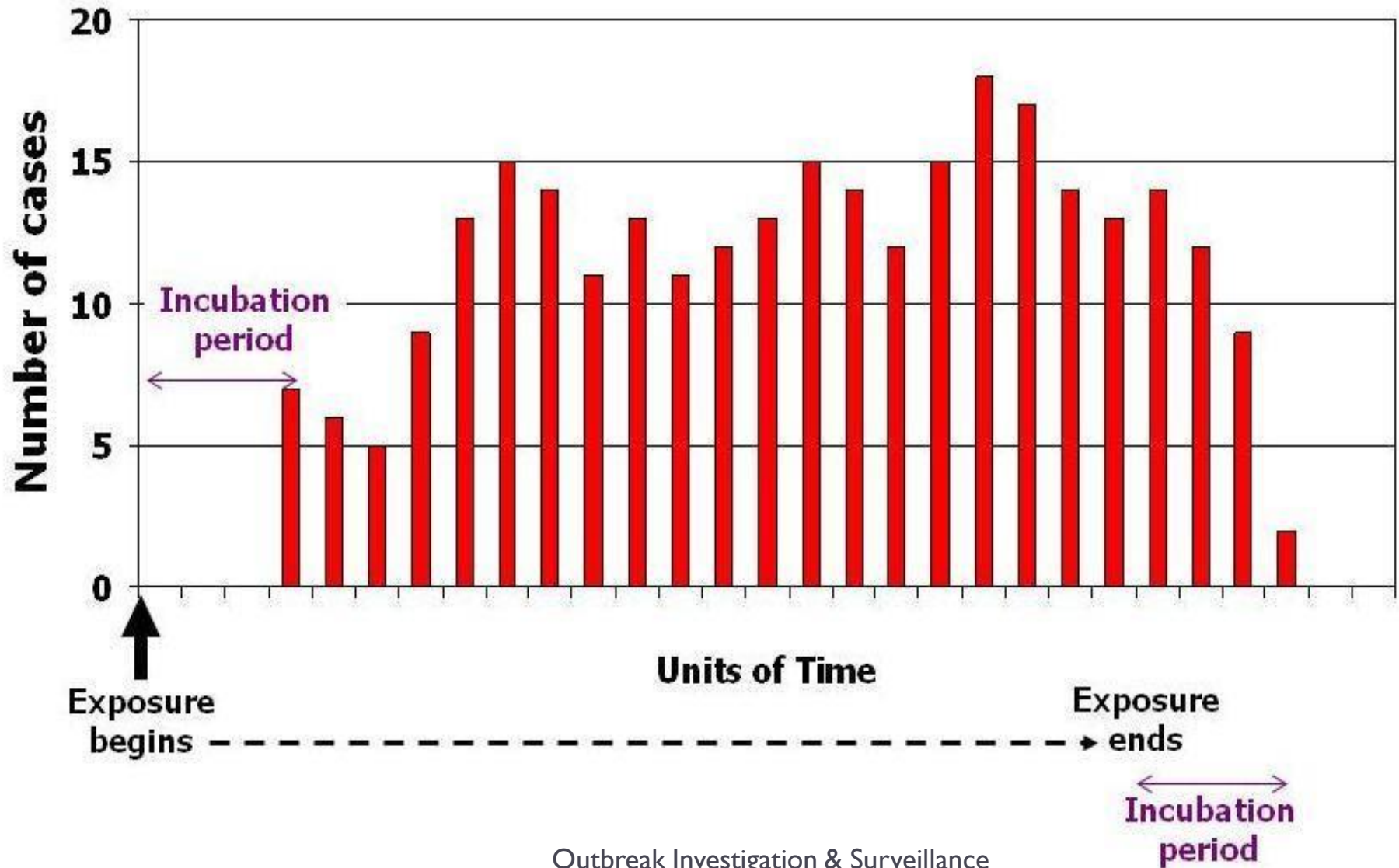
Point Source outbreak with no propagation



b) Continuing Common Source

- ▶ As with the point source outbreak, a group of people are exposed to a single noxious influence. But here the exposure continues over a longer time.
- ▶ E.g a contaminated water supply that doesn't get fixed, so the outbreak persists for longer. A recreational lake can be the source of leptospirosis to those who swim there.

Continuing Source Outbreak.

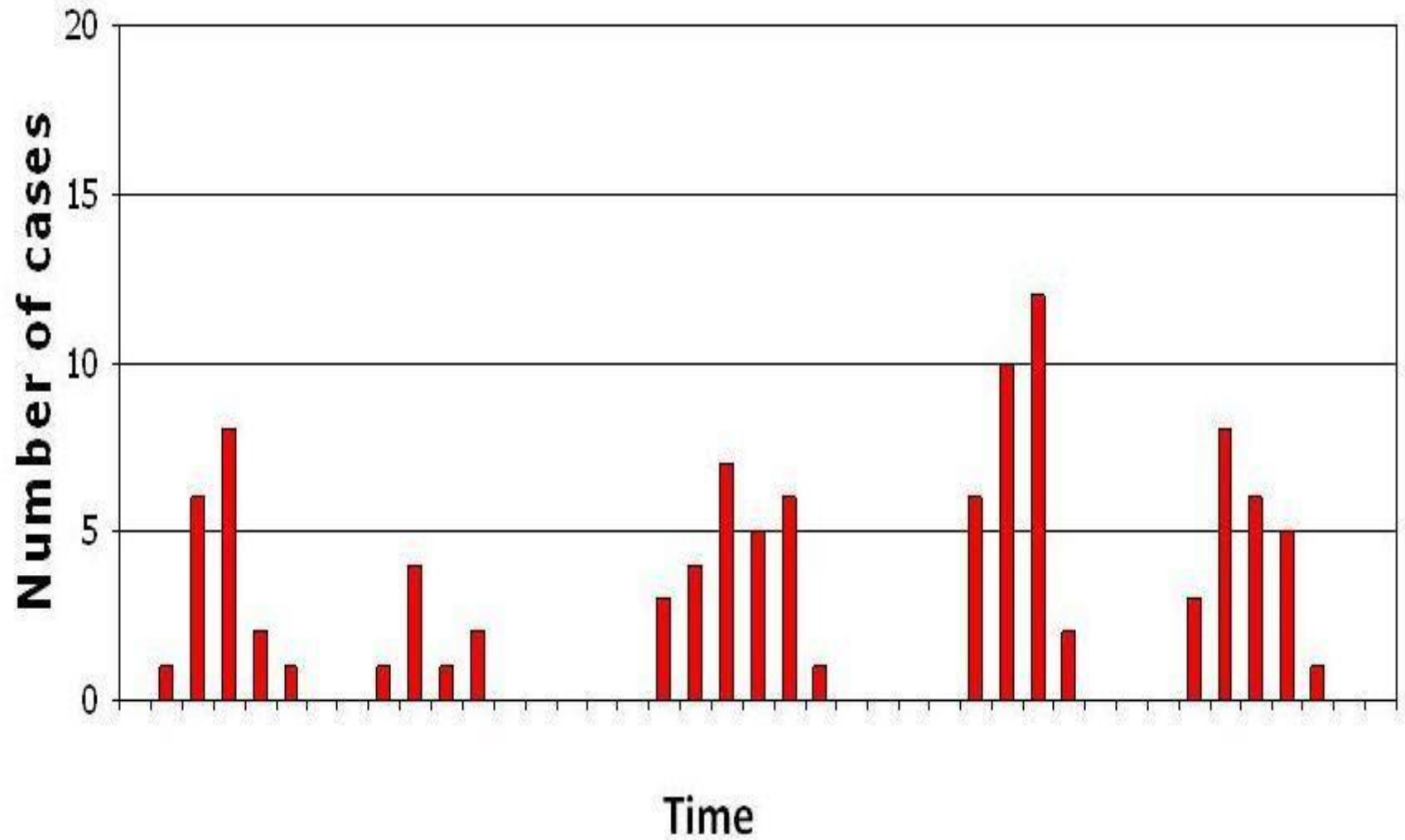


c) Intermittent Outbreaks

- ▶ You may occasionally see this pattern. This seems to be a common source that is not well controlled, so outbreaks reoccur.
- ▶ It could be seasonal or weather-related, or perhaps due to a common source such as an industrial contaminant being emitted at intervals.



Intermittent outbreak



II. Propagated Spread

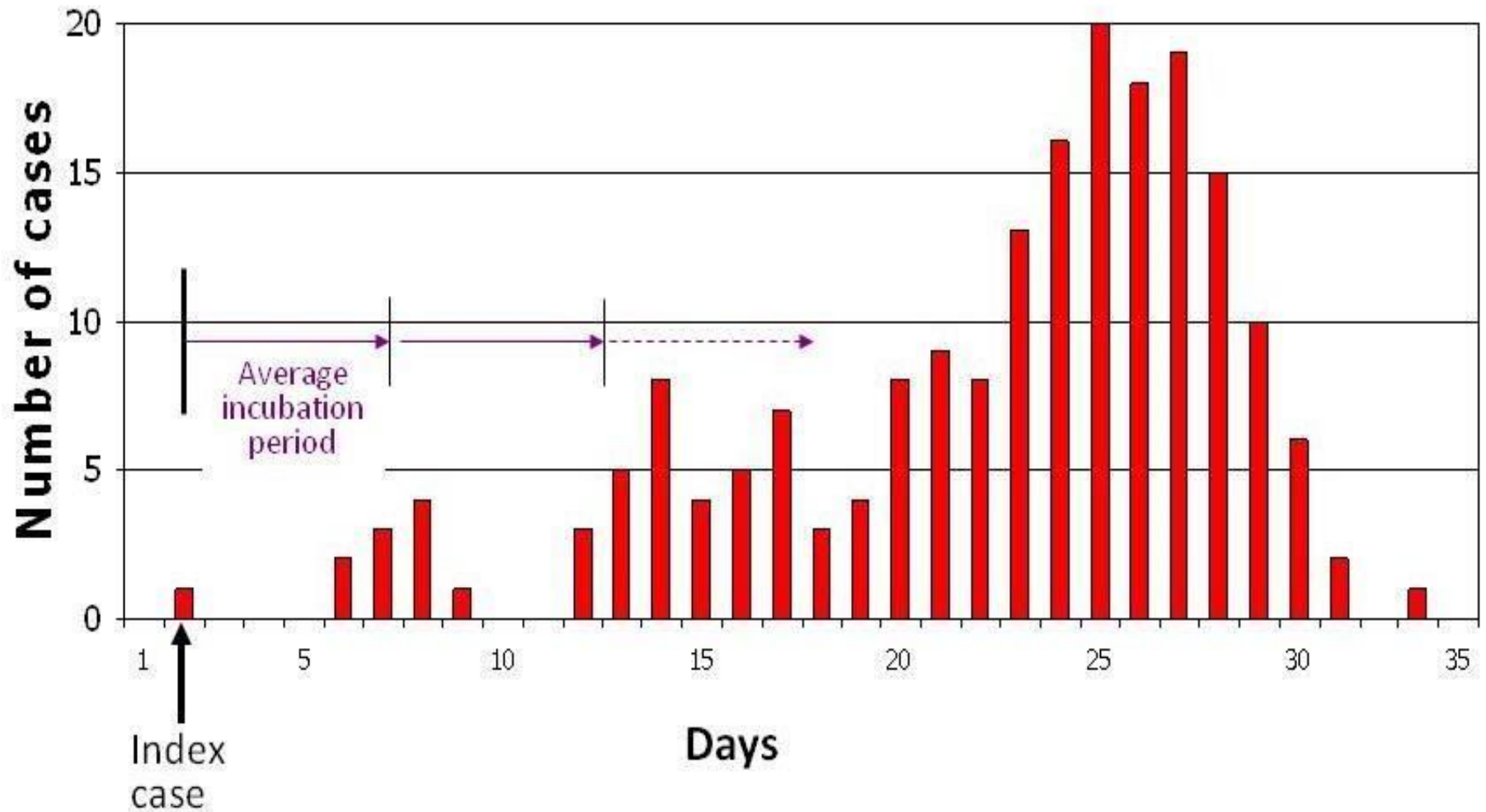
- ❑ Disease spreads via person-to-person contact.
- ❑ Here a single 'index' case (for example, a returning traveler) infects other people, and cases arise after an incubation period.
- ❑ Controlling the source is no longer sufficient to control the outbreak but successful control measures (isolation or quarantine) is required.



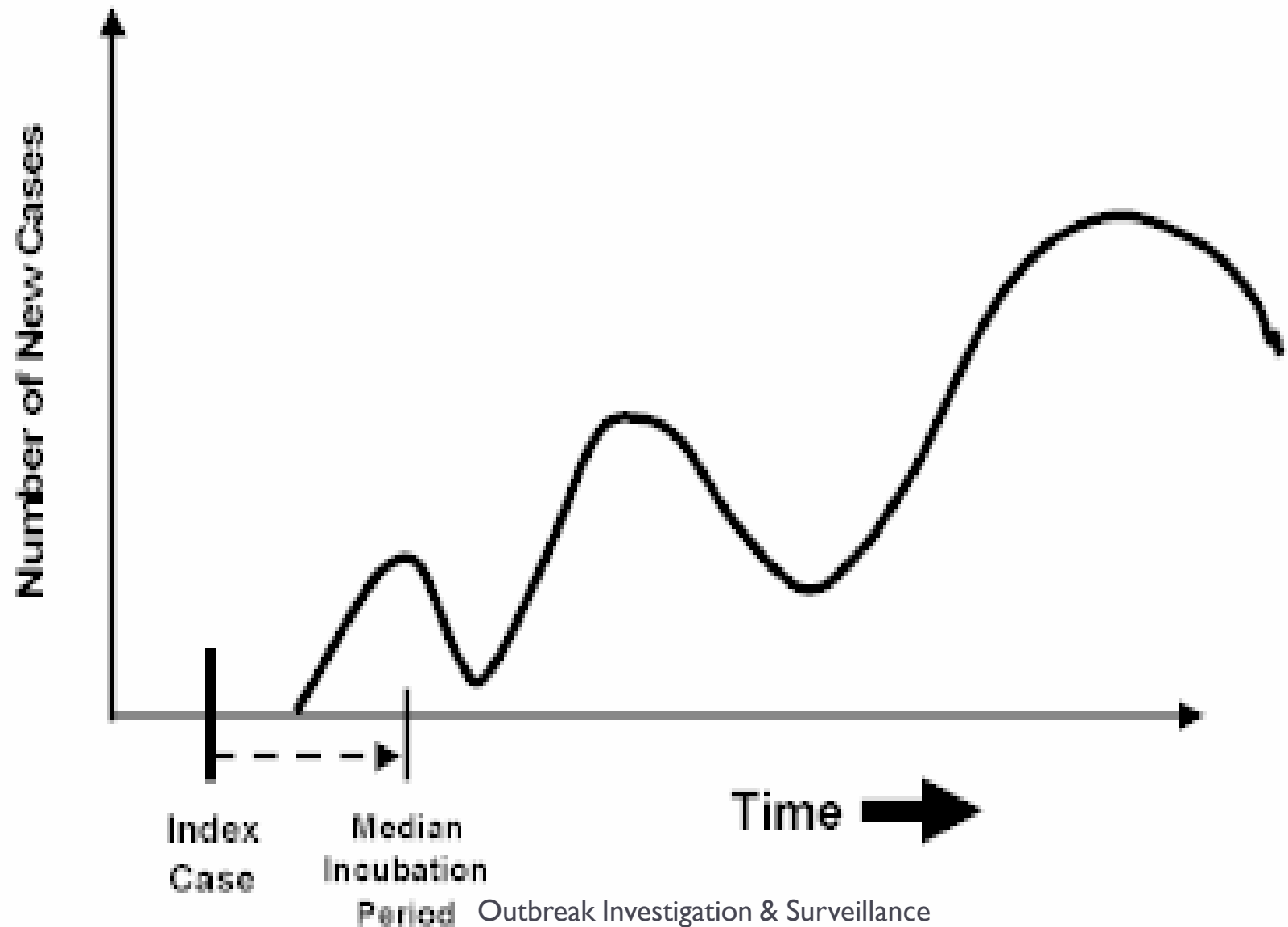
Con...

- **Quarantine-** is the limitation of freedom of movement of apparently healthy persons or animals who have been exposed to a case of infectious disease.
- **Cholera, Plaque, and yellow fever** are the three internationally quarantable diseases by international agreement.
- **Isolation:** (separation, segregation) denoting a hospital or ward for patients with contagious or infectious diseases.

Disseminated outbreak originating from an index case with propagated spread



Epidemic Curve of Propagating Epidemic



III. Mixed Epidemics

- ▶ Epidemics having the features of both common source and propagated epidemics are referred as mixed epidemics.
- ▶ A common source outbreak may be followed by secondary person-to-person spread.
- ▶ Example: Typhoid Fever

WHO Declares COVID-19 a Pandemic

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Affiliations  expand

PMID: 32191675 PMCID: [PMC7569573](#) DOI: [10.23750/abm.v91i1.9397](#)

Free PMC article

Abstract

The World Health Organization (WHO) on March 11, 2020, has declared the novel coronavirus (COVID-19) outbreak a global pandemic (1). At a news briefing, WHO Director-General, Dr. Tedros Adhanom Ghebreyesus, noted that over the past 2 weeks, the number of cases outside China increased 13-fold and the number of countries with cases increased threefold. Further increases are expected. He said that the WHO is "deeply concerned both by the alarming levels of spread and severity and by the alarming levels of inaction," and he called on countries to take action now to contain the virus. "We should double down," he said. "We should be more aggressive." [...].



Uncovering Outbreaks

- Observation of a single event or cluster of events by a clinician or laboratory personnel.
- Report of one or more patients
- Routine **surveillance** activities



Purposes of an Outbreak Investigation

- ▶ Stop the outbreak
 - ▶ Ensure public's health / Prevent spread of disease
 - ▶ Usually requires:
 - ▶ Identifying the agent, reservoir, source, and/or mode of transmission
 - ▶ Determining who is at risk for disease, place, time
 - ▶ Identifying the exposures or risk factors that increase risk of disease
- ▶ Prevent future outbreaks
- ▶ Improve surveillance and outbreak detection

Three-Pronged Approach

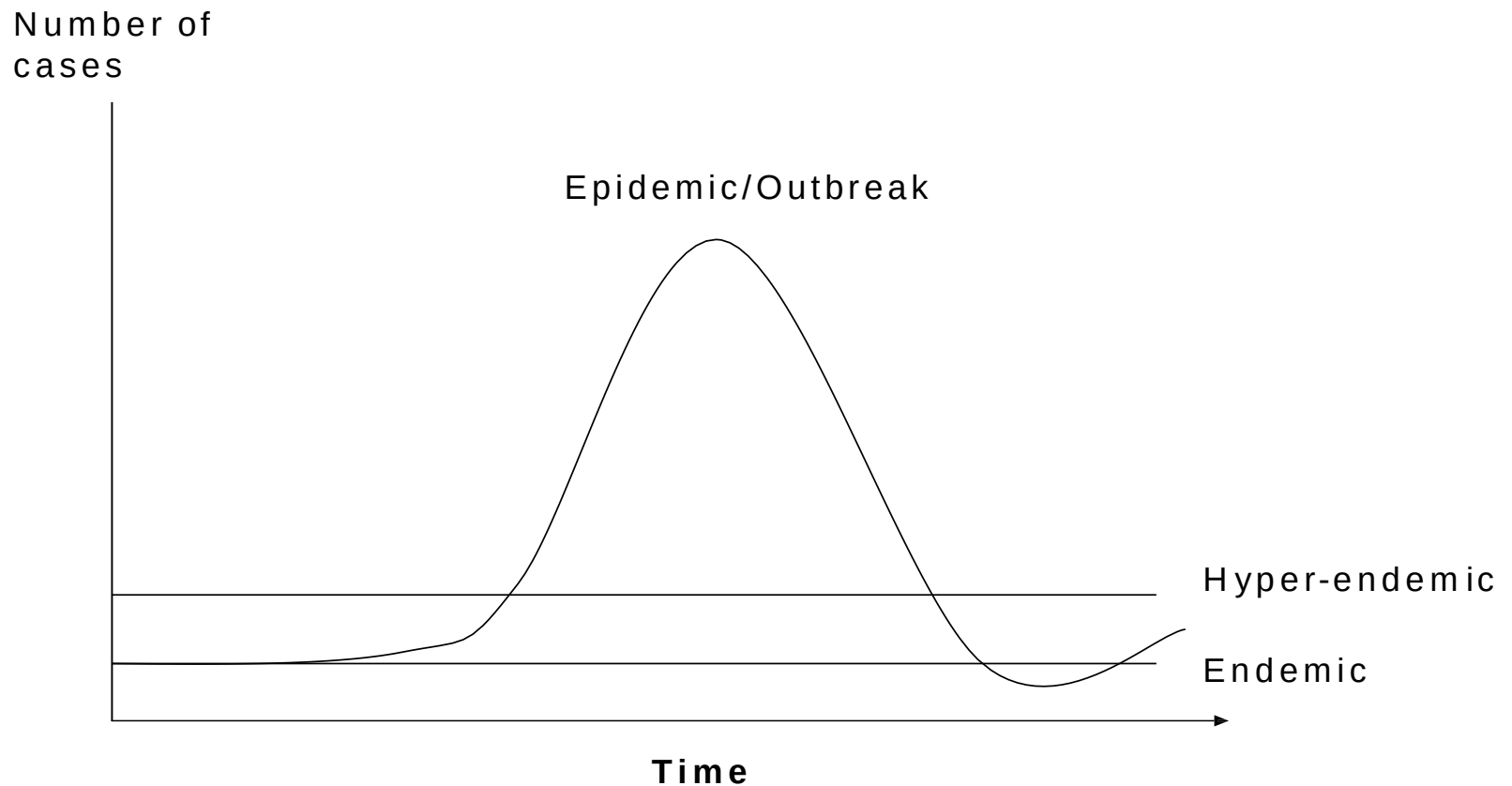
- ▶ Epidemiologic investigation
- ▶ Laboratory testing
- ▶ Environmental assessment



Steps of Outbreak Investigation

- 1. Confirm existence of an outbreak**
- 2. Verify the diagnosis**
 - < Make decision about conducting field investigation >
- 3. Prepare for field work**

Step 1: Confirm Existence of an Outbreak: Endemic versus Epidemic



Step 2: Verify the Diagnosis

- Ensure proper diagnosis
 - rule out laboratory error
- Examine / see case-patients
- Review medical records
- Confirm caseness



Step 3: Prepare for Field Work

- ✓ Investigation needs
 - Expertise
 - Sample questionnaires, supplies
 - Laboratory Equipment
 - Plan of action

- ✓ Management / operational issues
 - Team composition, role of each members
 - Communications plan
 - Where to go / whom to meet



Step 4a: Construct a Working Case Def

- ▶ **Case definition:-** is a standard set of criteria for deciding whether an individual should be classified as having the health condition of interest.
- ▶ Simple, Sensitive versus specific.
- ▶ Develop a case definition using:
 - ▶ Symptoms or laboratory results
 - ▶ Affected population (person)
 - ▶ Location (place)
 - ▶ Time frame



Categories of Case definition

- ▶ **Confirmed/definite:** a case with laboratory verification.
- ▶ **Probable :** a case with typical clinical features of the disease without laboratory confirmation.
- ▶ **Possible :** a case presented with fewer of the typical clinical features.

Example _Case Definition of COVID 19

Suspected case:

A patient with acute respiratory illness (**fever or history of fever AND at least one sign/symptom of respiratory disease** (e.g., cough, **shortness of breath**), regardless of a history of travel to or residence in a location reporting community transmission of COVID-19 disease

OR

A patient with acute respiratory illness (**fever or history of fever AND at least one sign/symptom of respiratory disease** (e.g., cough, **shortness of breath**), regardless of having contact history with a confirmed or probable **COVID-19 case**



Case Definition

Suspected case

A patient with severe acute respiratory illness (fever and at least one sign/symptom of respiratory disease, e.g., cough, shortness of breath, AND requiring hospitalization) AND in the absence of an alternative diagnosis that fully explains the clinical presentation.

Probable case:

A suspect case for whom testing for COVID-19 is inconclusive OR
If the suspect is not tested due to any reason

Confirmed case:

A person with laboratory confirmation of COVID-19 infection, irrespective of clinical signs and symptoms.



Sample will be collected if one or more of the following criteria fulfilled

1. If the rumor/alert is investigated and fulfill one or more of suspected case definition criteria, OR
2. If the individual/patient has a history of contact with confirmed or probable case regardless of sign and symptoms, OR
3. If there is an unusual increase or cluster of cases/deaths of respiratory illness in a specific area after investigated, OR



Step 4b: Find Cases Systematically and Develop Line List

- ▶ Collect data on case-patients using case def.
- ▶ Create line listing
- ▶ Line listing = rectangular database similar to spreadsheet
 - ▶ Each row represents one case
 - ▶ Each column represents one variable
 - ▶ First column usually identifier — name or ID number
 - ▶ Can be quickly reviewed and updated

Example of a Line Listing

		Signs/Symptoms			Labs	Demographics	
Case #	Date of Symptom Onset	Diarrhea	Vomiting	Fever >37°C	Positive stool culture	Age	Gender
1	22/10/05	Y	Y	Not done	Y	19	M
2	25/10/05	N	Y	N	N	17	M
3	22/10/05	N	Y	N	Y	23	F
4	27/10/05	Y	?	?	Pending	18	?
5	23/10/05	N	Y	N	Y	21	M
6	21/10/05	Y	Y	Y	Not submitted	18	F

Step 5. Tabulate Data by Time, Place, Person

- ▶ **Time** (*epidemic curve*)

- ▶ plots the cases by the time of onset and provides a time frame for the outbreak investigation.

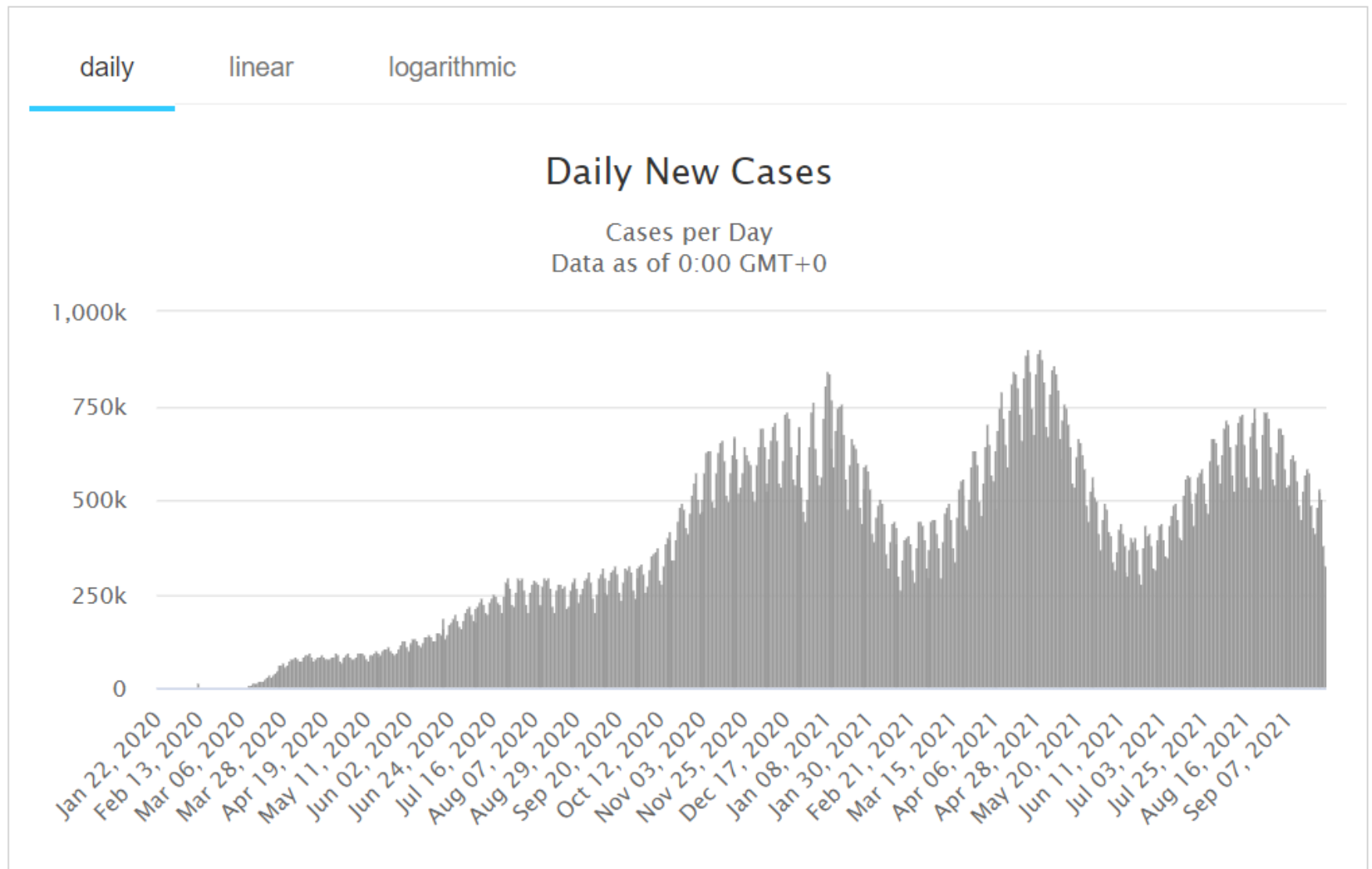
- ▶ **Place** (*spot map*)

- ▶ plots the cases by location and shows the geographic spread of cases.

- ▶ **Person** (*tables*)

- ▶ Who was infected?
- ▶ Calculate attack rate and secondary attack rate

Epi curve of COVID 19





340,845
confirmed cases

5,369
deaths

 Data may be incomplete for the current day or week.

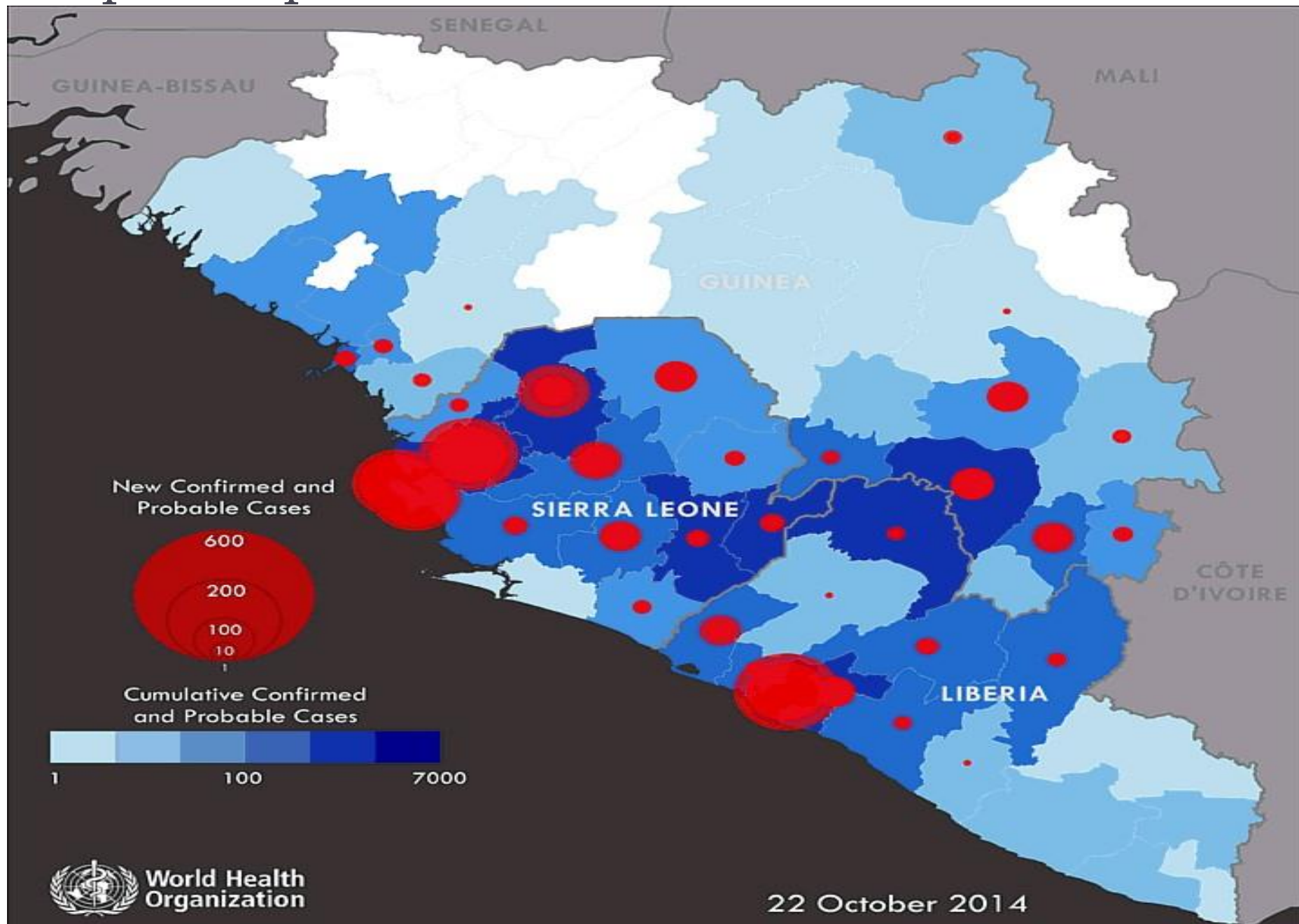
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Spot map of COVID_19

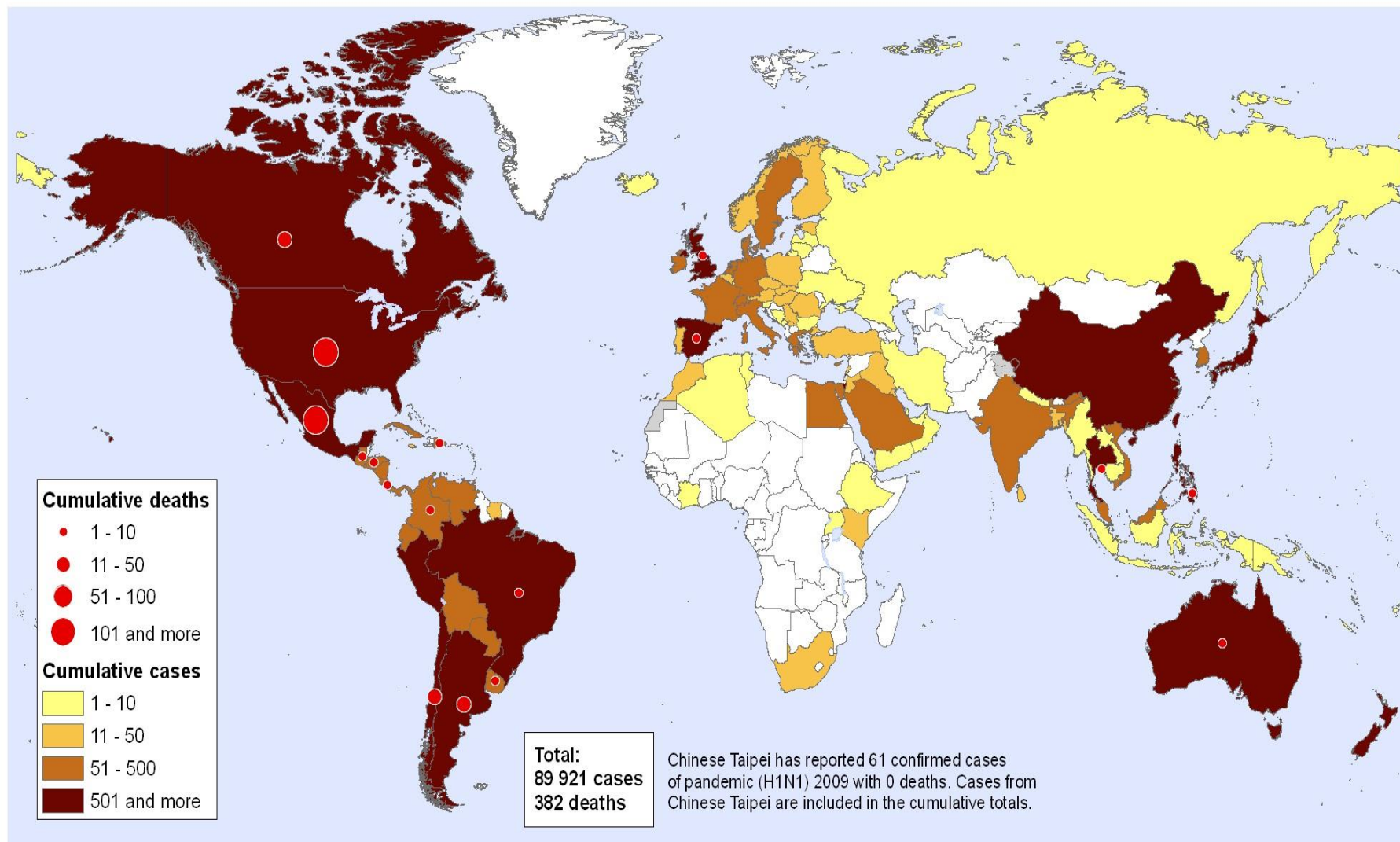


Spot Map of Ebola



**Pandemic (H1N1) 2009,
Number of laboratory confirmed cases and deaths as reported to WHO**

**Status as of 03 July 2009
09:00 GMT**



The boundaries and names shown and the designations used on this map do not imply the expression of any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted lines on maps represent approximate border lines for which there may not yet be full agreement.

Data Source: World Health Organization
Map Production: Public Health Information
and Geographic Information Systems (GIS)
World Health Organization



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Map produced: 03 July 2009 09:00 GMT

Step 6: Develop Hypotheses

- ▶ Review descriptive epidemiology
- ▶ Unique exposure opportunities
- ▶ Talk to case-patients – what do they think?
- ▶ What do local health officials think?

Step 7: Evaluate Hypotheses

- ❑ We need to analyze and test the hypothesis.



Step 8: Refine, Reconsider Hypotheses, ... Reconcile with Environmental and Lab Results

- **Conduct additional studies**
 - Environmental sampling
 - Laboratory-based

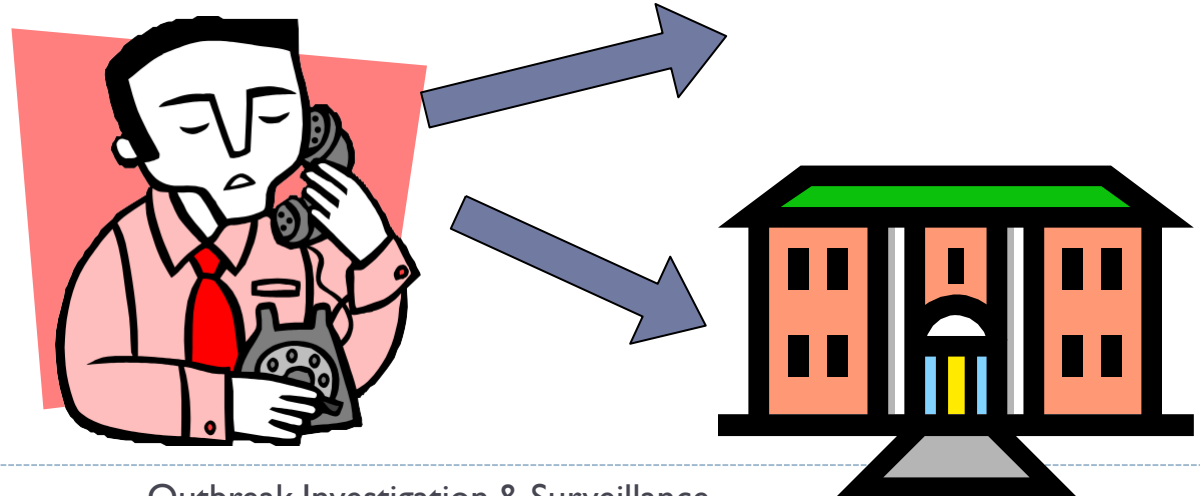
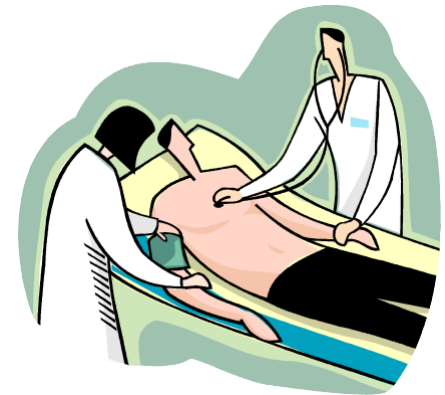


Step 9: Implement and evaluate control measures

- Implement as early as possible!
- Usually targeted to one or more:-
 - ▶ eliminate the source
 - ▶ interrupt transmission
 - ▶ reduce susceptibility
 - ▶ evaluate both short and long-term success.

Step 10: Communicate Findings

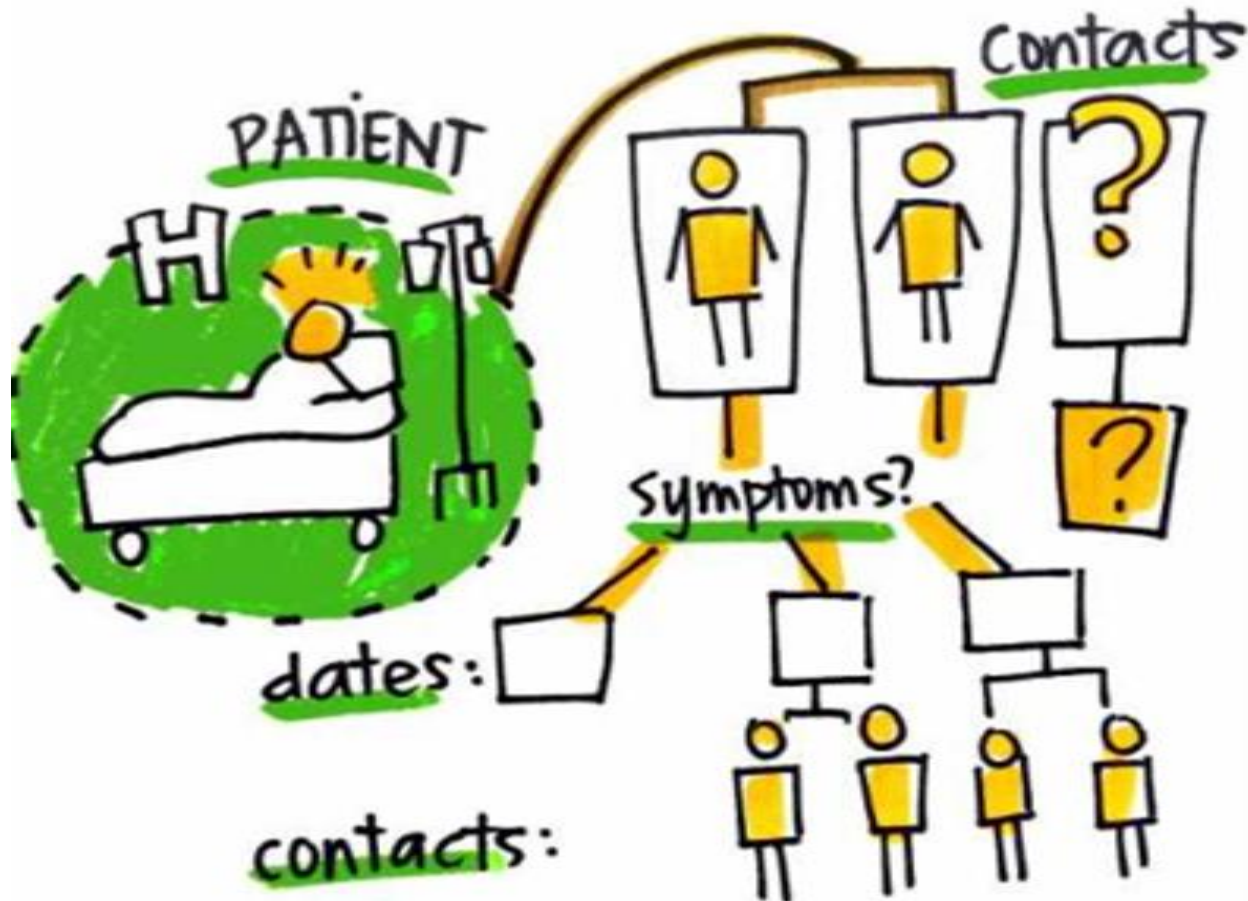
- ☐ Summarize investigation
- ☐ Make recommendations
- ☐ Disseminate report back to health care providers and other public health authorities





Surveillance

What do you understand from this picture?



Introduction –Surveillance

What is Surveillance?

- It is the *ongoing, systematic*:
 - ✓ collection, compilation, analysis, interpretation and dissemination /feedback to those who need to know, in order that action may be taken.

Surveillance is Information for Action

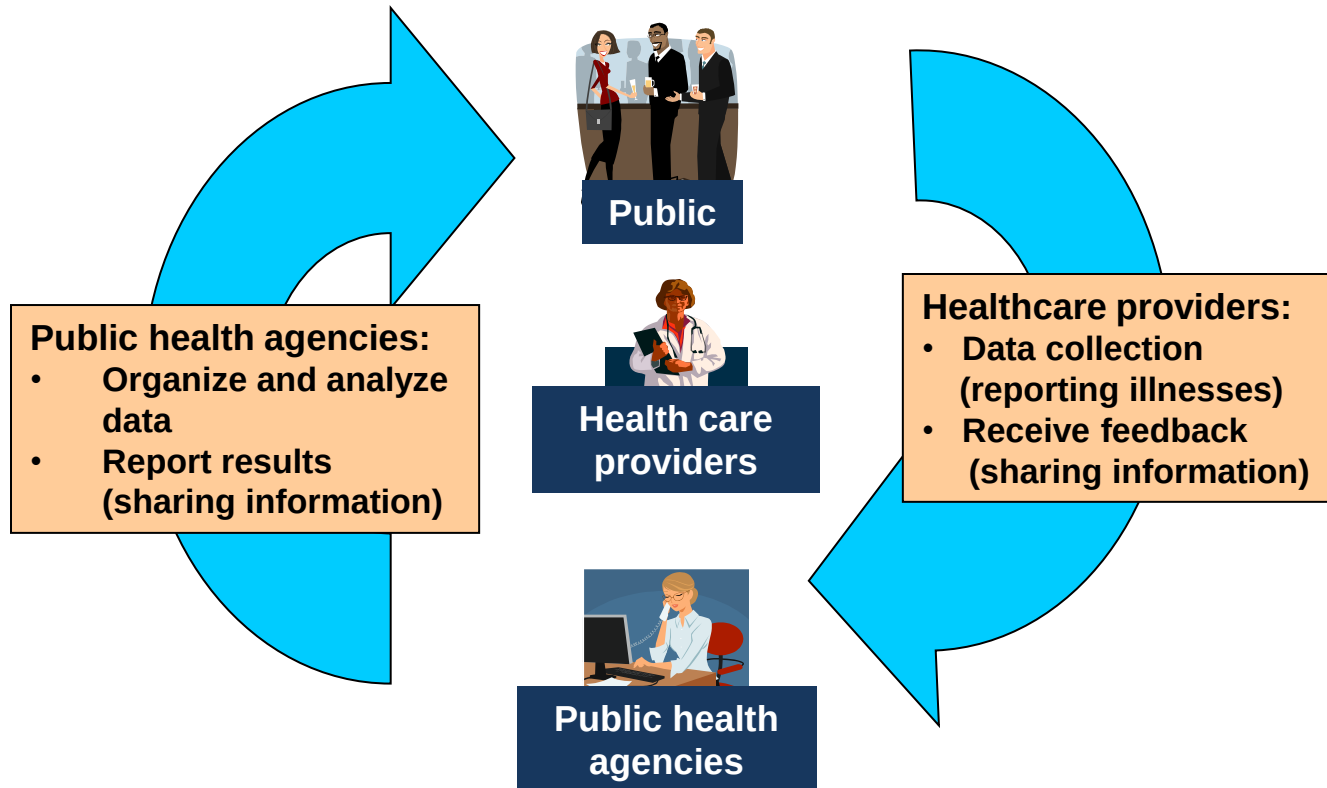


- 
- ✓Collection
 - ✓Analysis
 - ✓Interpretation
 - ✓Dissemination

Public Health Action

- ✓Planning
- ✓Monitoring and Evaluation
- ✓Policy making
- ✓Guide research
- ✓programs/interventions
- ✓Predict outbreaks/epidemics
- ✓Estimate future disease impact

Cycle of Public Health Surveillance



Purpose and Uses of Surveillance

- ▶ To detect sudden change in disease occurrence and distribution.
- ▶ To learn more about the natural history, clinical spectrum and epidemiology of a disease.
- ▶ To follow secular (long-term) trends of a disease.

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- ▶ To identify changes in agent, host and environment factors.
- ▶ To assess the effectiveness of a preventive intervention in a population.
- ▶ To generate (formulate) a hypothesis.



Criteria for inclusion in Surveillance

1. The current impact of the health event

- ▶ Having high incidence/ prevalence and Mortality, case fatality rate, Health care costs.

eg Malaria, Pneumonia, Diarrheal dis., TB, HIV/AIDS

2. Having epidemic potential

- ✓ (eg. Measles, cholera, meningitis)



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3. Surveillance required internationally

✓ (eg plague, yellow fever, cholera)

4. Having available and effective prevention and control interventions

✓ (eg EPI, NTD)

5. Can easily be identified using simple case definition



List of Priority Diseases in Ethiopia

Epidemic-prone diseases

1. Cholera
2. Diarrhea with blood (Shigellosis)
3. Measles
4. Meningitis
5. Plague
6. Viral hemorrhagic fever
7. Yellow fever



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- 8. Typhoid fever
- 9. Relapsing fever
- 10. Epidemic typhus
- 11. Malaria

Diseases targeted for eradication

- 12. Acute flaccid paralysis (Polio)
- 13. Dracunculiasis (Guinea worm)
- 14. Leprosy
- 15. Neonatal Tetanus



Other diseases of public health importance

- 16. Pneumonia in children
- 17. Diarrhea in children
- 18. New AIDS cases
- 19. Onchocerciasis
- 20. Sexually transmitted diseases
- 21. Tuberculosis
- 22. Rabies



Once diseases for surveillance are selected we should do the following items:

1. Prepare case definition of the diseases included in the surveillance.
2. Determining and updating population under surveillance. (Identifying vulnerable population)



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3. Setting the time period of collected data reporting (immediate, weekly, and monthly)

1. **Immediate reporting:**

- ☐ Presence of a single disease to result a suspicion for an Epidemic. Eg COVID, Polio, Cholera, Viral Hemorrhagic fever

2. **On weekly basis:**

- ☐ For epidemic prone diseases. Eg Malaria, meningitis

3. **On monthly basis:**

- ☐ For routine surveillance. Eg Tuberculosis, Leprosy, AIDS

4. Source of data, who will report etc.

5. How data are handled (confidentiality)



Case detection

Several actions need to happen concurrently to find additional cases and to prevent further spread, including:

- Actively find cases
- Exhaustive case and outbreak investigations
- Conduct enhanced surveillance



Case detection ...

Means of case detection

1. Alert/rumor investigation
2. Contact tracing
3. Screening



Case study 1: individual activity (10 mins)

Objective: To detect cases using appropriate methods

Scenario

- A hotel manager reported a client who had cough and fever to EPHI/PHEM using 8335 toll free. You are the expert assigned in the Emergency Operating Center (EOC) responding to his call.

Discussion Question

1. What additional info. do you need?
2. What will be the next steps and actions should be taken ?

Instruction:

- *Complete the case individually and report orally.*
- Time to complete the case 5 minutes and 5 minute for reporting



Contact tracing

- Objective of contact tracing is to enable early detection of cases through active monitoring of peoples exposed to possible risks and rapidly contain the outbreak.
- Active case finding involves a wider search, focusing on:
- Patients and their visitors in health care facilities where the confirmed patient seek for care
- Health care providers who cared for or cleaned the room of an infected patient
- Social, familial and work contacts of the infected patient
- Co-travelers of the infected patient
- Community



CLOSE Contact (High risk exposure)

A person:

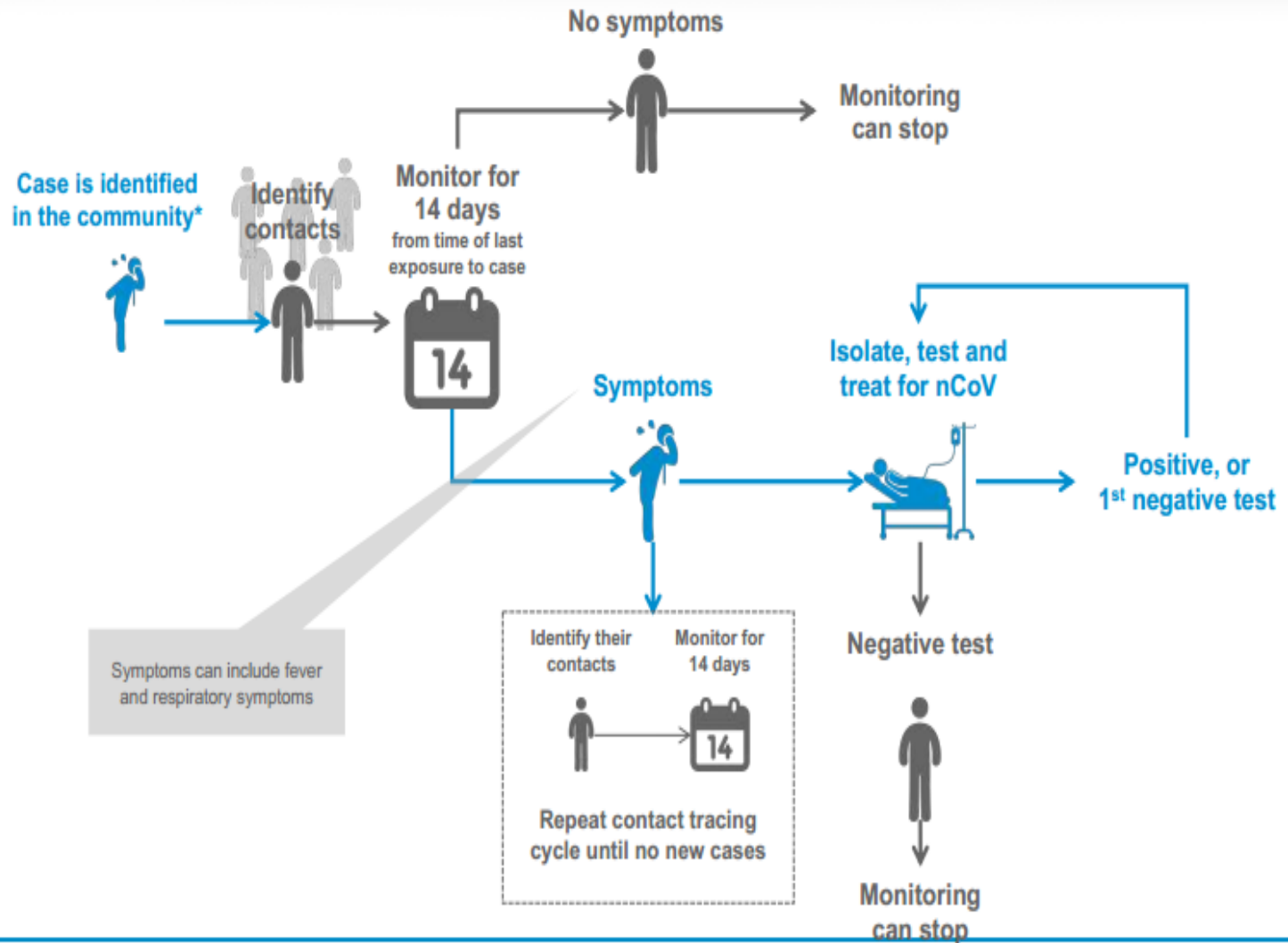
- Who had face-to-face contact with a COVID-19 case **within 1 metre for more than 15 minutes**
- Who had physical contact with a COVID-19 case
- Had unprotected direct contact with infectious secretions of a COVID-19 case (e.g. being coughed on)
- Who was in a closed environment (e.g. household, classroom, meeting room, hospital waiting room, etc.) with a COVID-19 case for more than 15 minutes
- In an aircraft, sitting within two seats (in any direction) of the COVID-19 case, travel companions or persons providing care, and crew members serving in the section of the aircraft where the index case was seated. (if severity of symptoms or movement of the case indicate more extensive exposure, passengers seated in the entire section or all passengers on the aircraft may be considered close contacts)
- A healthcare worker or other person providing care to a COVID-19 case, or laboratory workers handling specimens from a COVID-19 case, without recommended PPE or with a possible breach of PPE

CASUAL Contact (Low risk exposure)

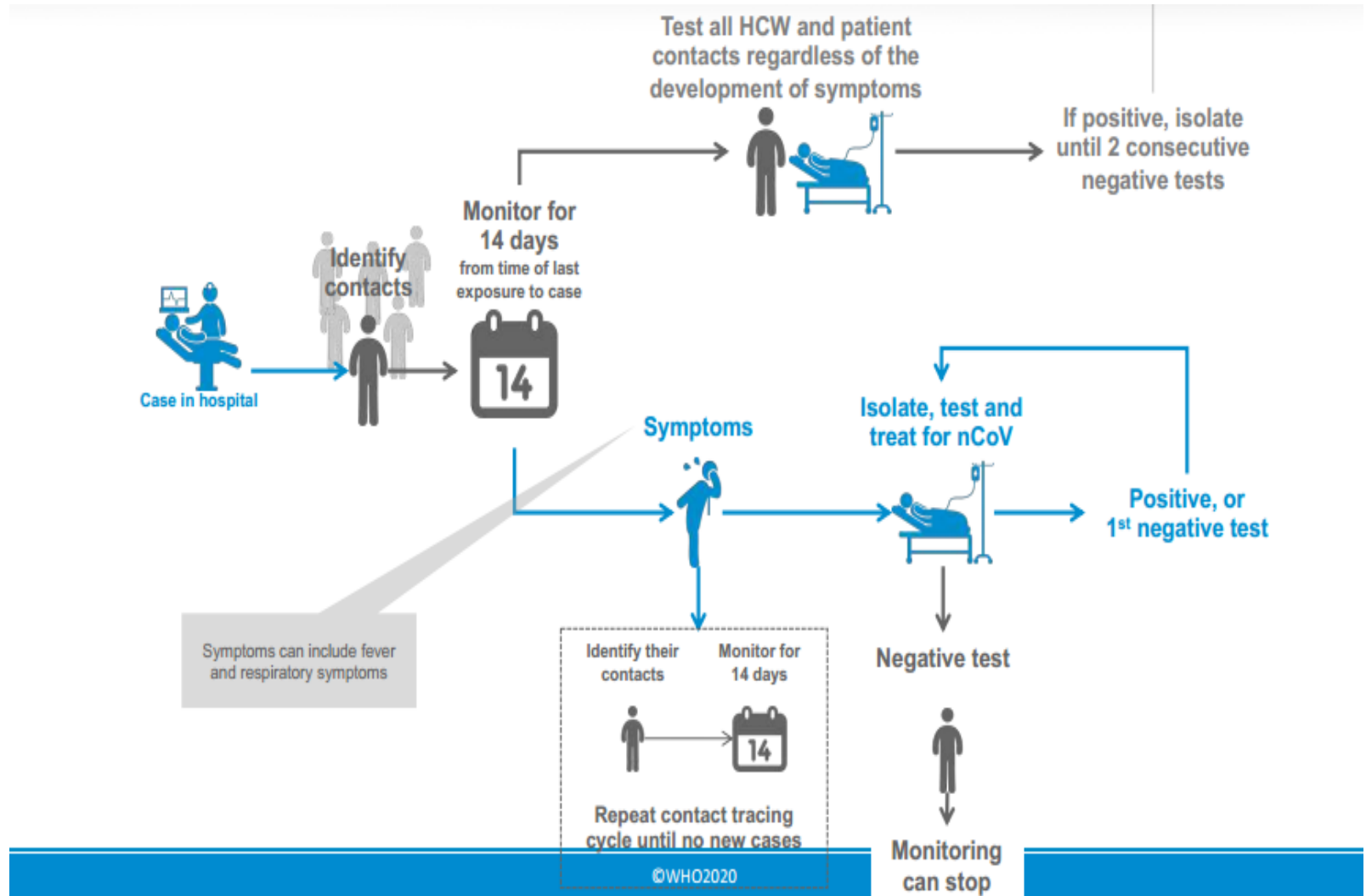
A person:

- Who had face-to-face contact with a COVID-19 case **within 1 metre for less than 15 minutes**
- Who was in a close environment with a COVID-19 case for less than 15 minutes
- Travelling together with a COVID-19 case in any mode of transport
- A healthcare worker or other person providing care to a COVID-19 case, or laboratory workers handling specimens from a COVID-19 case, wearing the recommended PPE

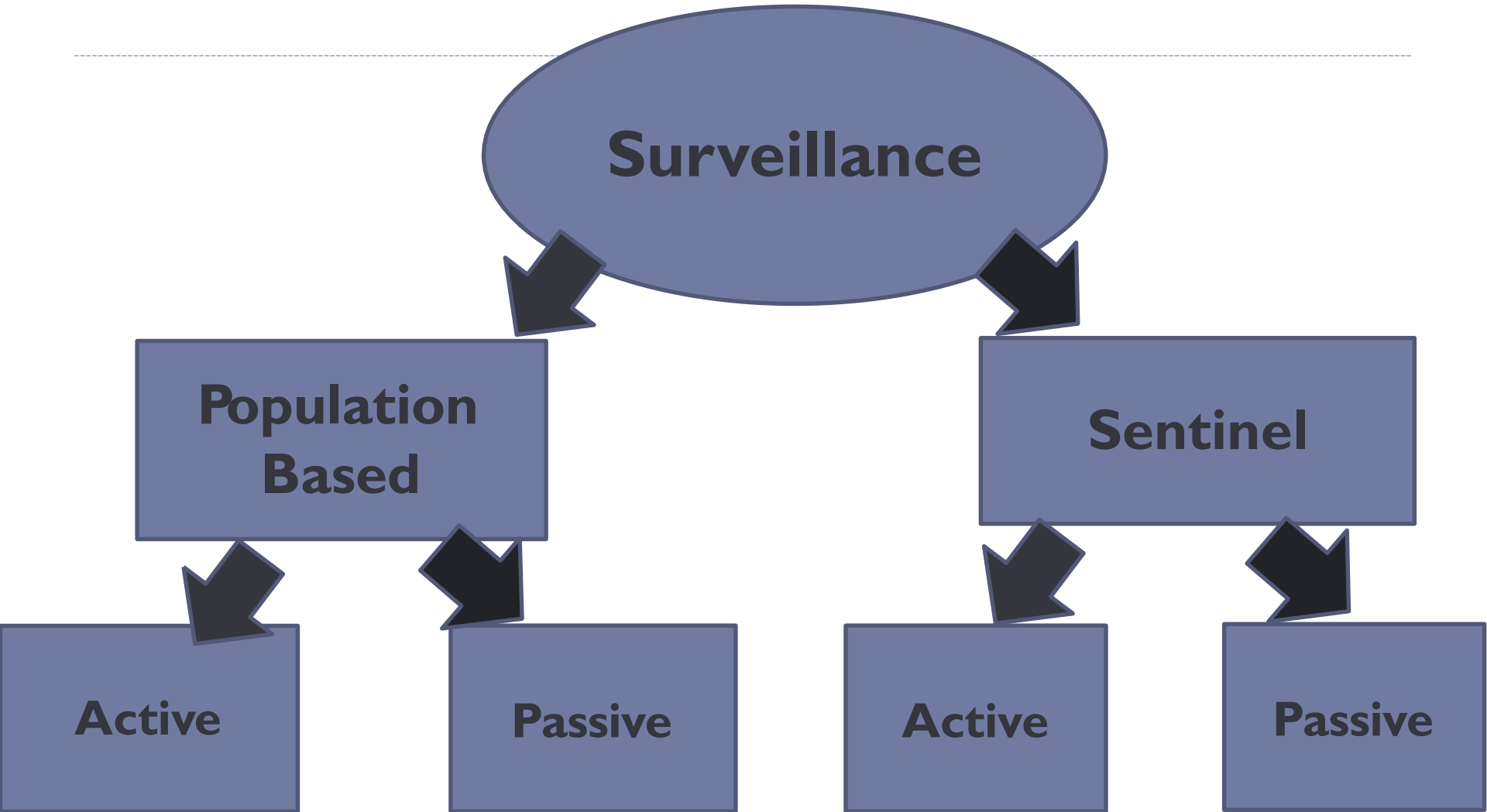
Contact tracing when case is identified in community



Contact tracing when case is at health facility



Types of Surveillance



Active and Passive

- ▶ **Passive** – provider initiated reporting
(routine reporting)
- ▶ **Active** – public health system initiated reporting, or
(Public health official) initiated through resource
mobilization.

Active Systems: Characteristics

- ▶ Assure more complete reporting
- ▶ More representative
- ▶ Usually done for brief periods
- ▶ Requires more health department resources

Passive Systems: Characteristics

- ▶ Simple
- ▶ May not be representative
- ▶ May not be timely
- ▶ May fail to identify outbreaks

Population based and sentinel

- ▶ **Population based** :Surveillance pertaining to the general population defined by geographical boundaries.
- ▶ **Characteristics**
 - ▶ Relatively representative
 - ▶ Potential for detection of rare diseases



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-
- ▶ **Sentinel:** It is a surveillance based on selected population samples chosen to represent the relevant experience of particular groups.

Characteristics

- ☐ Selected physicians or facilities involved
- ☐ Specified diseases reported
- ☐ High quality data collected
- ☐ Lack of representativeness

Enhanced surveillance

- The aim is to limit human to human transmission through early detection
- **Where: Example for COVID_19**
 - At point of entry
 - Health Facilities (public and private)
 - Influenza hospital-based sentinel sites for SARI
 - Community



Features of good surveillance system

- ▶ Uses a combination of passive and active mechanisms to collect data.
- ▶ Collection of minimum data in a simplest possible way.
- ▶ To assure quality, it is good if the data collected is useful for the data collectors.
- ▶ Timely reporting and comprehensive action.

Evaluation of Surveillance

1. It's public health importance
2. The objectives and operation of the system.
(Case definition, reporting time, data handing...)
3. The system's usefulness
4. Attributes or qualities of the surveillance system (simplicity, flexibility, acceptability, representativeness, timeliness)

Thank you!

Any question - abitowon@gmail.com